

## 1. MSE Function + Logistic Sigmoid Function Issue:

A. in the output layer, we can focus on single output perceptron  $R$  for a pair training example,

$$\Rightarrow \text{Error}_R = \frac{\partial e}{\partial z_R} = \frac{\partial e}{\partial f_R} \cdot \frac{\partial f_R}{\partial z_R}$$

$$\Rightarrow \hat{y} = f_R = S(z_R)$$

$$\Rightarrow \text{Error}_R = e'(\hat{y}) \cdot S'(z_R) = -\frac{1}{2}(\hat{y} - y) \cdot S(z_R) \cdot (1 - S(z_R))$$

$$\Rightarrow \text{Assume } y = 0, \text{ but } \hat{y} \text{ computed} = 0.99,$$

$$\Rightarrow \text{Error}_R = -\frac{1}{2}(0 - 0.99) \cdot 0.99 \cdot (1 - 0.99) = 0.0049$$

B. the loss between ground truth ( $y=0$ ) and predicted output ( $\hat{y}=0.99$ ) is completely reverse,

however, the value of  $\text{Error}_R$  (0.0049) approach 0 after implementing a series of derivative,

in this case, loss cannot be backward passed smoothly, and then the weights can't be adjusted correctly finally.

## 2. Error Function — Cross Entropy:

$$e(\hat{y}) = -\frac{1}{m} \sum_{i=0}^{m-1} (y_i \cdot \ln(\hat{y}_i) + (1 - y_i) \cdot \ln(1 - \hat{y}_i)), \left[ \begin{array}{l} y_i = \text{desire output } y \text{ (ground truth) from } i\text{-th pair training example which corresponding to single output perceptron} \\ \hat{y}_i = \text{predicted output } \hat{y} \text{ from single output perceptron which corresponding to } i\text{-th pair training example} \\ m = \text{number of pair training examples} \end{array} \right]$$

## 3. Cross Entropy Function + Logistic Sigmoid Function Solution:

A. in the output layer, we can focus on single output perceptron  $R$  for a pair training example,

$$\Rightarrow \text{Error}_R = \frac{\partial e}{\partial z_R} = \frac{\partial e}{\partial f_R} \cdot \frac{\partial f_R}{\partial z_R}$$

$$\Rightarrow \hat{y} = f_R = S(z_R)$$

$$\Rightarrow \text{Error}_R = e'(\hat{y}) \cdot S'(z_R) = -\left(\frac{y}{\hat{y}} + \frac{1-y}{1-\hat{y}}\right) \cdot S(z_R) \cdot (1 - S(z_R))$$

$$\Rightarrow \text{Assume } y = 0, \text{ but } \hat{y} \text{ computed} = 0.99,$$

$$\Rightarrow \text{Error}_R = -\left(\frac{0}{0.99} + \frac{1-0}{1-0.99}\right) \cdot 0.99 \cdot (1 - 0.99) = -0.99$$

B. the loss between ground truth ( $y=0$ ) and predicted output ( $\hat{y}=0.99$ ) is completely reverse,

therefore, the value of  $\text{Error}_R$  (-0.99) doesn't approach 0 after implementing a series of derivative,

in this case, loss can be backward passed smoothly, and then the weights can be adjusted properly finally.

## 4. Cross Entropy Function + Hyperbolic Tangent Function Limitation:

A. in the output layer, we can focus on single output perceptron  $R$  for a pair training example,

$$\Rightarrow \text{Error}_R = \frac{\partial e}{\partial z_R} = \frac{\partial e}{\partial f_R} \cdot \frac{\partial f_R}{\partial z_R}$$

$$\Rightarrow \hat{y} = f_R = \tanh(z_R)$$

$$\Rightarrow \text{Error}_R = e'(\hat{y}) \cdot \tanh'(z_R) = (y \cdot \ln(\hat{y}) + (1 - y) \cdot \ln(1 - \hat{y}))' \cdot \tanh'(z_R)$$

B. Because the range of hyperbolic tangent function value is from -1 to 1,

it is undefined when  $\hat{y} = \tanh(z_R)$  is negative and is put into  $\ln(\hat{y})$ ,

in this case, we won't use the set combining with cross entropy function and hyperbolic tangent function.

## 5. Cross Entropy Function + Logistic Sigmoid Function Stability:

A. in the output layer, we can focus on single output perceptron  $R$  for a pair training example,

$$\Rightarrow \text{Error}_R = \frac{\partial e}{\partial z_R} = \frac{\partial e}{\partial f_R} \cdot \frac{\partial f_R}{\partial z_R}$$

$$\Rightarrow \hat{y} = f_R = S(z_R)$$

$$\Rightarrow \text{Error}_R = e'(\hat{y}) \cdot S'(z_R) = -\left(\frac{y}{\hat{y}} + \frac{1-y}{1-\hat{y}}\right) \cdot S(z_R) \cdot (1 - S(z_R))$$

$$\Rightarrow \text{Assume } y = 0, \text{ but } \hat{y} \text{ } z_R = 40$$

$$\Rightarrow \hat{y} = S(40) = \frac{1}{1 + e^{-40}} \approx \frac{1}{1 + 10^{-18}} \rightarrow \frac{1}{1+0} = 1,$$

$\because 10^{-18}$  underflows the lower bound of +floating point number as the graph below,

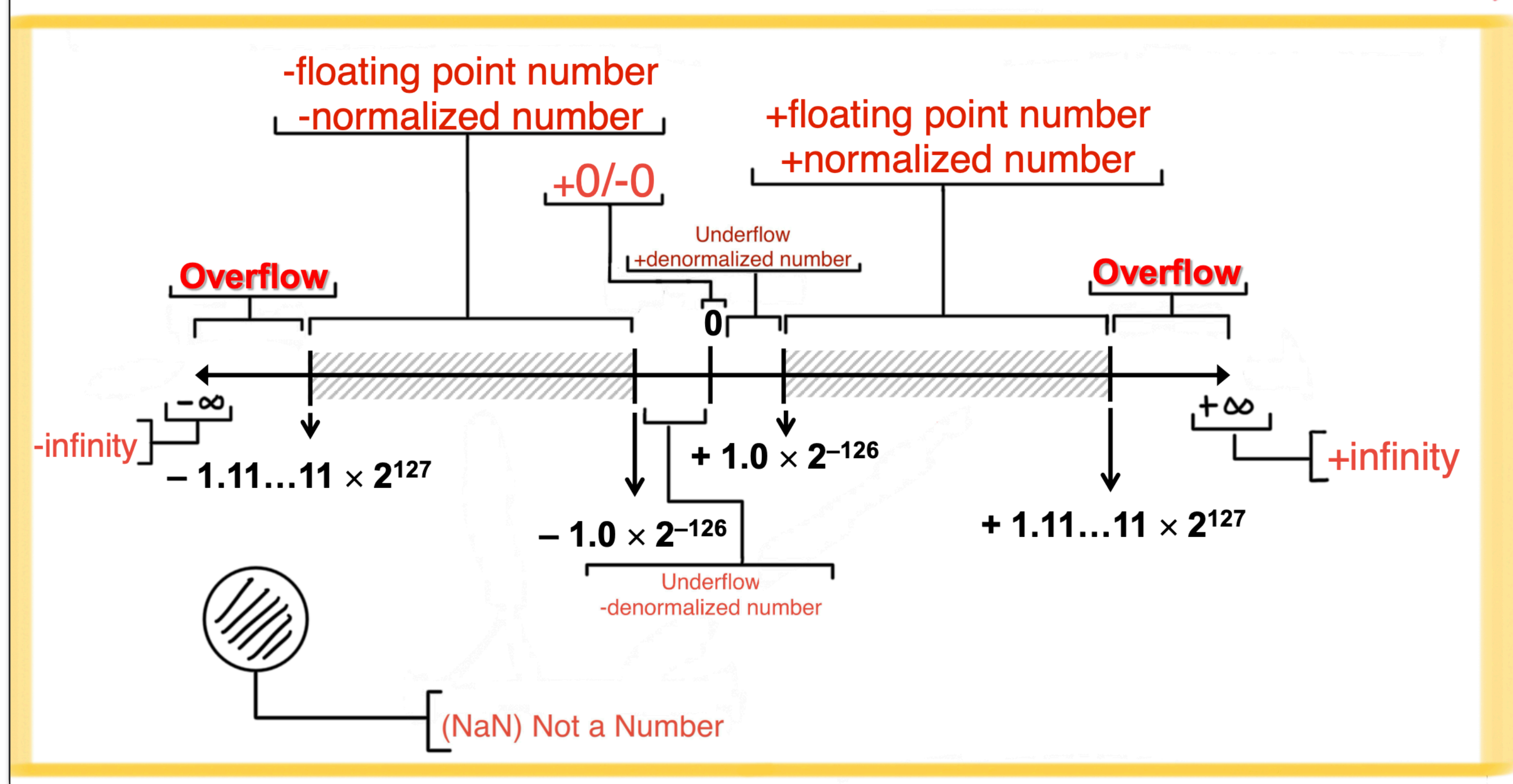
$$\Rightarrow \text{Error}_R = -\left(\frac{0}{1} + \frac{1-0}{1-1}\right) \cdot 1 \cdot (1-1) = -\text{inf} \cdot 0 \rightarrow \text{NaN}$$

$-\infty \rightarrow -\text{inf}$ ,  $\because -\infty$  overflows the lower bound of -floating point number as the graph below,

$\because -\text{inf} \cdot 0$  can't be defined as the graph below,

$\Rightarrow$  ☆ ☆ ☆

## Range of Single-Precision Floating Point Number



B. to avoid the results  $\text{NaN}$ , we compute the formula of  $\text{Error}_R = e'(\hat{y}) \cdot S'(z_R) = -\left(\frac{y}{\hat{y}} + \frac{1-y}{1-\hat{y}}\right) \cdot S(z_R) \cdot (1 - S(z_R))$  into  $\text{Error}_R = \hat{y} - y$  first,

and then we calculate the value of  $\text{Error}_R$  with real number ( $\text{Error}_R = \hat{y} - y = 1 - 0 = 1$ ),

in this case, we can get  $\text{Error}_R = 1$  instead of  $\text{NaN}$ .